

Proposed Revision of a Class-AB Analog Pad Driver in SG13CMOS5L: Matched-Pair DDA, MOS Miller Capacitors, and a Resized Front End

C. H. Maier

Abstract—The first version of a differential-to-single-ended class-AB pad driver in IHP’s SG13CMOS5L used the pad’s ESD clamp frames as output devices. Its area was dominated by two MOM Miller capacitors (1922 of $2387\mu\text{m}^2$ outside the frames). Its gain was set by a ratio of two degenerated pairs, $(R_2 + 2/g_m)/(R_1 + 2/g_m)$, so it moved by $-2.6\ldots + 5.0\%$ over corners. Its offset came from the folding sinks, and its loop gain was limited by short internal cascodes. The poly resistors themselves occupy $28\mu\text{m}^2$. The revision keeps the frames and the floating class-AB control. It replaces the differential difference amplifier (DDA) with four identical units balanced as $f(v_{\text{inp}} - v_{\text{ref}}) + f(v_{\text{ref}} - v_{\text{inn}}) = 2f(v_{\text{fb}} - v_{\text{ref}})$, which sets the gain $\frac{1}{2}$ by the unit count and cancels the units’ nonlinearity as far as they match: a static “externally linear, internally nonlinear” circuit. The Miller capacitors become thick-oxide PMOS accumulation capacitors, and the front end and the internal cascodes are resized. Into $1\text{k}\Omega \parallel 100\text{pF}$ the supply current drops from 427 to $296\mu\text{A}$, the loop gain rises from 66.5 to 78.9 dB, and the gain over 11 corners stays within $0.4992\ldots 0.5000$. The DC nonlinearity drops from 5.5 to 1.2 mV, THD at 10 kHz from -51 to -58.5 dB, the mismatch offset σ from 14.4 to 5.3 mV, and the output noise at 1 kHz from 1.6 to $0.68\mu\text{V}/\sqrt{\text{Hz}}$. Three resistor-free degenerations from the literature were checked: a triode MOS, a well input, and an undegenerated square-law pair. None fits this topology, because each unit’s output must depend on the gate difference alone. The output devices cannot be cascoded while they are the ESD clamps. Moved into the slot and cascoded, they give a capacitor-free, load-compensated driver with 74 dB of loop gain, for capacitive loads only.

Index Terms—Class-AB output stage, differential difference amplifier, externally linear circuits, ESD, matched-nonlinearity cancellation, MOS capacitor, Miller compensation, pad driver, SG13CMOS5L.

I. WHERE THE FIRST VERSION’S NUMBERS COME FROM

TABLE I traces each figure of merit of the first version [1] to its cause. Only one of the causes is a linear passive: the MOM capacitors. The resistors set the gain error, but through their combination with g_m , not by being resistors.

The offset budget follows Sarpeshkar’s rule [2, §3.3.4]: the input offset is the current mismatch divided by the transconductance,

$$V_{os} = \Delta I / G_m, \quad (1)$$

Design note in the IEEE journal format; companion to class_ab_pad_driver.md. Simulation only (ngspice-42, IHP-Open-PDK dev, tt, 27°C unless stated), no layout, not reviewed. Drafted with Claude (configured model claude-opus-5-5; the serving model may differ), 2 October 2026. Running log: log.md; sources: literature_notes.md; every number: sim/results_improvements.txt, produced by sim/run_improvements.py.

TABLE I
FIRST VERSION: FIGURES OF MERIT AND THEIR CAUSES

quantity	value	traced to
area outside frames	$2387\mu\text{m}^2$ (gate 437, MOM 1922, rhigh 28)	MOM Miller capacitors on M1–M4
supply current	$427\mu\text{A}$	output I_Q 240, bias fixture ≈ 90 , front end ≈ 100
gain over corners	$0.4870\ldots 0.5249$	$(R_2 + 2/g_m)/(R_1 + 2/g_m)$: R tracks R , g_m does not
DC nonlinearity	5.5 mV (12.7 mV at ss 27°C)	degenerated pairs, outside the loop
offset σ (30 seeds)	14.4 mV	sinks 11.5, tails 4.2, mirror 2.3, pairs 1.7 mV ^a
loop gain, $1\text{k}\Omega \parallel 100\text{pF}$	66.5 dB	A_1 (short cascodes) $\times$ $A_2 = g_{m,\text{out}}(R_L \parallel r_o) \approx 6$

^aBudget of the MP-DDA in the first version’s sizing (fold, sinks and mirror identical), one device group at a time, 20 seeds.

so a front end with a low G_m/I is offset-prone, and the mirrors and sinks matter more than the input devices.

II. MATCHED-PAIR DDA

A. Principle

Four identical transconductance units each deliver $f(g_p - g_n)$ to the fold nodes: unit A with $(g_p, g_n) = (v_{\text{inp}}, v_{\text{ref}})$, unit B with $(v_{\text{ref}}, v_{\text{inn}})$, and units C1 and C2, both with $(v_{\text{ref}}, v_{\text{fb}})$, subtracting. The loop forces

$$f(v_{\text{inp}} - v_{\text{ref}}) + f(v_{\text{ref}} - v_{\text{inn}}) = 2f(v_{\text{fb}} - v_{\text{ref}}). \quad (2)$$

With the input common mode at v_{ref} , $v_{\text{inp}} - v_{\text{ref}} = v_{\text{ref}} - v_{\text{inn}} = u$, and (2) reads $2f(u) = 2f(v_{\text{out}} - v_{\text{ref}})$. Hence $v_{\text{out}} - v_{\text{ref}} = (v_{\text{inp}} - v_{\text{inn}})/2$ for any odd, monotonic f . For an input common-mode error c , the input units carry

$$f(u + c) + f(u - c) = 2f(u) + f''(u)c^2 + O(c^4), \quad (3)$$

so the error, $f''(u)c^2/2f'(u)$, is second order in c . The gain $\frac{1}{2}$ is the ratio 2:4 of unit counts. The input units compress, and the feedback units apply the inverse of the same law. This is the static form of ELIN: Minch’s “function inversion by negative feedback around a high-gain amplifier” [3], and the cancellation the Säckinger–Guggenbühl DDA [16] shows when both pairs carry the same signal.

B. Requirement on a Unit

One input of every unit sits at v_{ref} . For the same differential voltage x , the input common mode, and with it the source

voltages, moves by $+x/2$ in A and C and by $-x/2$ in B. So a unit qualifies only if its output depends on $g_p - g_n$ alone, not on where its sources sit; its own linearity is secondary. The test bench sweeps $f_A(x) = I(v_{\text{ref}} + x, v_{\text{ref}})$ and $f_B(x) = I(v_{\text{ref}}, v_{\text{ref}} - x)$ and solves $f_A(u) + f_B(u) = 2f_A(v)$ (Table II). For the resistor unit this gives $-1.04/-1.02$ mV at $u = \mp 0.5$ V. The full driver's largest deviation from its small-signal line is 1.04 mV.

The split-tail resistor unit qualifies because each input device is a constant-current source follower. A follower's transfer does not depend on the device law [3, §7.2.1], so the element between the sources sees exactly $g_p - g_n$. The other three fail for one common reason. A triode MOS's resistance depends on its V_{SG} , which is the absolute source voltage. The well input's κ depends on the well-to-gate voltage [2, §3.3.1]. A single-tail pair's tail and channel-length modulation see the source move. The design rule is therefore “no element whose value depends on the source voltage”, not “no resistor”, and the poly resistor stays. A unit cannot be made symmetric in common mode here, because the feedback side has only v_{fb} and v_{ref} ; that would need an inverted replica of the output.

C. Result With Nothing Else Changed

d2s_mp keeps the first version's fold, mirror, class-AB control, frames, MOM capacitors and bias, and the same 80 μA of tail current as $8 \times 10 \mu\text{A}$ (Table III). The gain σ is now the rhigh unit mismatch: 1.0 % for $0.5 \times 36.4 \mu\text{m}$ units and 0.84 % for the $0.5 \times 50.6 \mu\text{m}$ units of d2s_mpdpa. Four times the resistor area would halve it, at about 75 μm^2 more per unit. The offset is untouched, because it lives in the fold (Section IV).

III. MILLER COMPENSATION WITHOUT BEOL CAPACITORS

A. Device

Each Miller capacitor is an sg13_hv_pmos used as an accumulation capacitor, with the gate above the n-well. On the A side the gate is node a (≈ 2.6 V) and source, drain and well are v_{out} . On the B side the gate is v_{out} and source, drain and well are node b (≈ 0.7 V). Over the linear output range V_{GW} spans 0.15...1.15 V (A) and 0.25...1.25 V (B). $C_{ox} \approx 5 \text{ fF}/\mu\text{m}^2$ ($t_{ox} = 6.9 \text{ nm}$), and in that range the device delivers 0.38...0.76 of $C_{ox}A$. A $14 \times 14 \mu\text{m}$ device has 0.370 pF at 0.15 V, 0.648 pF at 1.0 V and 0.745 pF at 1.25 V; a $31 \times 31 \mu\text{m}$ MOM capacitor has 1.002 pF.

B. Equivalence

$18 \times 18 \mu\text{m}$ per side reproduces the MOM pair's loop within a degree (Table IV). That is 648 μm^2 of gate oxide instead of 1922 μm^2 of M1–M4 fingers. THD with capacitive loads is unchanged ($-60.0/-60.0$ dB at 100 pF). With $1 \text{ k}\Omega \parallel 100 \text{ pF}$, 100 kHz THD loses 2.3 dB (-53.2 vs. -55.5 dB): the C–V curvature modulates f_c where the loop gain is lowest.

C. Why the Capacitor Cannot Simply Be Made Small

With Miller compensation the non-dominant pole is

$$\omega_{p2} \approx \frac{g_{m,\text{out}} C_M}{C_L (C_g + C_M)}. \quad (4)$$

The frame gates carry $C_g \approx 0.86 \text{ pF}$ (P) and 0.69 pF (N), so pole splitting needs C_M of the order of C_g whatever G_{m1} is; lowering G_{m1} only lowers f_c . Two ways around this floor were considered but not simulated:

- Source-follower buffers in front of the frame gates. They shrink the high-impedance node, so C_M could drop to perhaps 0.2 pF (estimate), but the class-AB loop would have to include the followers.
- Indirect (current-buffer) compensation [20] into the fold nodes. These sit at ≈ 0.37 V, which biases a MOS capacitor poorly. The a -side current would also have the wrong sign through the mirror unless injected at the mirror cascode's source.

Load compensation needs no capacitor at all, but only once the output can be cascoded (Section V).

D. ESD Check Item

The A-side capacitor puts a p⁺/n-well junction, the n-well/p-substrate diode and 7 nm of gate oxide between the pad and node a . The B-side capacitor puts gate oxide between the pad and node b . The frames' own gate–drain overlaps already carry the same oxide stress, but the capacitors belong in the ESD review.

IV. CURRENTS, OFFSET AND NOISE: D2S_MPDPA

Headroom fixes the sizing. Nodes a (≈ 2.6 V) and b (≈ 0.7 V) are the frame gates, which leaves about 0.35 V per device for the mirror and its cascode and for the sinks and theirs. The overdrive of the matching-critical devices cannot be raised, so their area is raised instead (Table VII). All conducting devices are at least 80 mV clear of saturation at $v_d = 0$. Table VIII compares the result with the first version.

The offset budget of d2s_mpdpa (σ , one group at a time, 20 seeds) is: sinks 4.3 mV, unit pairs 1.7 mV, tails 1.4 mV, mirror 1.4 mV, the rest below 0.1 mV. The sinks still lead.

The halved front-end G_m lowers f_c from 1.62 to 1.37 MHz ($1 \text{ k}\Omega \parallel 100 \text{ pF}$). That costs the 100 kHz THD gain of Section II and makes 50 Ω slightly worse. $14 \times 14 \mu\text{m}$ capacitors raise f_c to 2.12 MHz at 100 pF but leave 54° of phase margin; the choice belongs to the specification. 50 Ω remains what the first note called it: a DC and low-frequency case.

V. CASCODING

A. Output Devices Are the Pad's ESD Clamps

A series cascode would put its own drain, not the clamp's, on the pad. A stacked clamp frame would be a new ESD structure needing its own qualification. The output therefore stays uncascoded, and the frames' r_o ($V_A \approx 26$ V for P; $L = 1.0 \mu\text{m}$ for N) bounds the second stage. What can be cascoded are the internal high-impedance nodes a and b (Table IX). Regulated (gain-boosted) cascodes do not fit: the ≈ 0.35 V per device next to a and b leaves no room for an auxiliary amplifier's V_{GS} .

TABLE II
DDA UNITS: COMMON-MODE SENSITIVITY AND STATIC MP-DDA ERROR

unit	G_{m0}	G_m vs. input CM	own compression at 0.5 V	DDA error at $u = -0.5 / +0.5$ V
unit_r: split tails, rhigh between the sources	15.6 μ S	0.6 %/V	0.5 %	-1.0 / -1.0 mV
unit_t: same, triode hv-PMOS instead of rhigh	15.5 μ S	77 %/V	20 %	-191 / -72 mV
unit_w: well (bulk) input, gates at v_{ss} [2]	11.0 μ S	-27 %/V	6.8 %	+29 / +38 mV
unit_q: undegenerated square-law pair, 2/6 μ m	7.8 μ S	-72 %/V	22 %	+75 / +200 mV

TABLE III
MP-DDA IN THE FIRST VERSION'S SIZING

	first version	MP-DDA
gain, tt	0.5005	0.4998
gain, 11 corners	0.4870...0.5249	0.4983...0.4998
nonlinearity ± 1 V, tt / worst	5.52 / 12.69 mV	1.04 / 1.91 mV
THD 10/100 kHz,	-51.1 / -50.2 dB	-60.2 / -55.5 dB
1 k Ω 100 pF		
THD 10/100 kHz, 100 pF	-51.0 / -51.0 dB	-60.0 / -59.8 dB
T_0 (1 k Ω 100 pF)	66.5 dB	66.5 dB
f_c / PM (1 k Ω 100 pF)	1.62 MHz / 74.8°	1.62 MHz / 74.7°
gain σ / offset σ (30 seeds)	1.60 % / 14.4 mV	1.33 % / 14.5 mV

TABLE IV
CROSSOVER FREQUENCY / PHASE MARGIN VS. LOAD CAPACITANCE (NO RESISTIVE LOAD)

C_L	MOM 31 $\times$ 31 μ m	MOS 18 $\times$ 18 μ m
5 pF	2.18 MHz / 86.1°	2.14 MHz / 86.4°
20 pF	2.17 MHz / 82.6°	2.14 MHz / 82.9°
100 pF	2.03 MHz / 66.2°	2.00 MHz / 67.0°
300 pF	1.60 MHz / 45.5°	1.59 MHz / 46.3°
1 nF	1.00 MHz / 26.4°	1.00 MHz / 26.9°

B. Output Devices in the Slot

With the pad reached through `IOPadAnalog`, the output devices are ordinary devices and can be cascoded. `d2s_1c2` is load-compensated and has no compensation capacitor. It uses the MP-DDA front end and half-frame output devices, with the cascode gates at v_{ref} (output range $\approx 1.0 \dots 2.4$ V); see Table X. Cascoding adds 37 dB. It helps neither resistive loads, which collapse any single-stage loop, nor the minimum load capacitance ($\approx 30\text{--}40$ pF).

In `IOPadAnalog`, `padres` sits behind the $586.9\ \Omega$ of `SecondaryProtection`. Sensing the loop at `padres` (the near side) turns that resistor into an isolation resistor. The phase margin rises from 39° to 62° at 20 pF and reaches 104° at 100 pF; the pad then sees a $587\ \Omega \cdot C_L$ low-pass outside the loop, 2.7 MHz at 100 pF. Sensing at `pad` (the far side) keeps the margins of the bare stage. 50 Ω cannot be driven through 587 Ω .

Two points remain open. First, the cascoded version has 7.8 mV of systematic offset, because the diode replicas sit at a different V_{DS} than the cascoded output devices; cascoding the replicas should remove it. Second, an in-slot output stage is not an area saving. This is an estimate, not simulated: for the same output pole at the same current, in-slot devices for the two-stage topology would need about the frames' W , because the frames' $g_m/I \approx 12.5\ \text{V}^{-1}$ at 240 μ A is what sets the non-dominant pole.

VI. EXTERNALLY LINEAR CIRCUITS WITH A NON-EXPONENTIAL INTERNAL LAW

Edwards [8, p. 72] (see also [9], [10]) names the square law of strong inversion [15] and the exponential as the two nonlinearities that ELIN circuits can exploit. Table V lists what else the sources offer and how each fares here. Two expressions recur in it. Sarpeshkar's square-law identity [2, p. 25] is

$$(x - a)^2 - (x - b)^2 = (b - a)(2x - a - b). \quad (5)$$

For a push-pull stage with constant $V_{GS,P} + V_{GS,N} = 2V_Q$ and drive $\pm\delta$, it gives

$$I_P - I_N = 2\beta(V_Q - V_t)\delta, \quad |I_{out}| < 4I_Q, \quad (6)$$

exactly, in strong inversion.

ELIN without the exponential is therefore feasible here in one form: function inversion by matched units. Its precondition is not a particular law but a unit whose output is independent of the source voltage, which the source-follower-plus-resistor unit provides. The square-law translinear classics need either strong inversion at currents this driver does not spend, or balanced signals that a single-ended pad does not have.

VII. OPTIONS

Table VI compares the variants. Area is drawn device area (W·L, resistor w·l, capacitor w·l) without the frames and the bias fixture, as in the first note. The area-lean variant's corner gain is that of `d2s_mp`, since the capacitors do not enter the DC gain.

VIII. NOT DONE

- `xschem` schematics and the CACE run for `d2s_mpd`. The existing `yaml` should take it as a new DUT, with `d2s_bias_lp` as fixture.
- Layout. The four units must match, so common-centroid placement and Edwards's "same surround" rule [8, §3.10] apply; the PMOS input devices sit in separate wells (bulk at their own source).
- PSRR, CMRR, input CM range, supply corners (3.0/3.6 V) and process Monte Carlo (`tt_stat`). Only mismatch Monte Carlo was run.
- ESD review of the pad-connected MOS capacitors (Section III).
- `d2s_1c2`: cascoded replicas, a real minimum- C_L specification, and whether a slot-side driver is wanted at all.
- Inherited from the first version: the PMOS of the left floating replica (XFPL) carries 1 nA, because with its bulk at v_{dd} , $|V_{GS}| = 0.78$ V is below its body-shifted $|V_{th}|$ of

TABLE V
ELIN APPROACHES WITH NON-EXPONENTIAL INTERNAL LAWS: FEASIBILITY FOR THIS DRIVER

approach	internal law	sources	verdict
matched-nonlinearity feedback, $f^{-1} \circ f$	any odd monotonic f (here a degenerated square law)	[3], [16]	used ; the unit must depend on $g_p - g_n$ only (Section II)
MOS translinear loop	square law	[15], [17]	already present: the floating class-AB control [18], [19] is one
constant-sum square-law push-pull	square law	[2]	the frames run at $g_m/I \approx 12.5 \text{ V}^{-1}$ (moderate inversion); an exact square law would need mA of I_Q ; not pursued
$\sqrt{\cdot}$ -domain / dynamic translinear	square law or exponential	[11]–[14]	dynamic (filters); a static buffer has nothing to compand
triode MOSFET-C	triode, even orders cancel	—	needs balanced signals; the feedback side here is single-ended
capacitive dividers: FG input, MITE, AFGA, MIFG	capacitive + exponential or law-independent follower	[3]–[7]	needs linear capacitors, charge control (tunneling, injection or UV) and AC coupling; none is available or acceptable here
well input, bump linearization	κ -scaled law; bump in subthreshold	[2]	G_m depends on input CM: +29/ +38 mV error (unit_w); bump range is subthreshold-sized
undegenerated strong-inversion pair	square law	—	linear enough, but the sources move with the signal: +75/ +200 mV error (unit_q)

TABLE VI
VARIANTS COMPARED (1 k Ω || 100 pF UNLESS STATED)

variant	area	I_{DD}	offset σ	T_0 : 1 k Ω 100 pF / 50 Ω / 100 pF	gain over corners	THD 10/100 kHz
first version (d2s_miller)	2387 μm^2	427 μA	14.4 mV	66.5/40.7/96.5 dB	−2.6... +5.0 %	−51.1/ − 50.2 dB
area-lean: d2s_mp + MOS 18×18	1238 μm^2	427 μA	14.5 mV	66.5/40.7/96.5 dB	−0.3... 0 %	−60.1/ − 53.2 dB
d2s_mpdda, $L_{cas} = 1 \mu\text{m}$	1694 μm^2	296 μA	5.3 mV ^a	69.7/43.9/99.6 dB	not simulated	−58.7/ − 51.4 dB ^a
d2s_mpdda, $L_{cas} = 3 \mu\text{m}$ (default)	2014 μm^2	296 μA	5.3 mV	78.9/53.1/104.3 dB	−0.16... 0 %	−58.5/ − 51.3 dB

^aFrom the intermediate run in log.md (d2s_mp2: the circuit of d2s_mpdda with $L_{cas} = 1 \mu\text{m}$ except $n_g = 1$ in the cascodes; its loop numbers agree with the $L_{cas} = 1$ row of results_improvements.txt), not from that file itself.

TABLE VII
SIZING CHANGES

block	first version	d2s_mpdda
unit tails	2 pairs $\times 2 \times 20 \mu\text{A} = 80 \mu\text{A}$	4 units $\times 2 \times 5 \mu\text{A} = 40 \mu\text{A}$ (5/6 μm)
degeneration	rhig 116 k Ω / 51 k Ω	rhig 150 k Ω per unit ($I_t R = 0.75 \text{ V}$)
folding sinks	$2 \times 50 \mu\text{A}$, 25/2	$2 \times 30 \mu\text{A}$, 36/6
mirror, cascodes	10/2, $L = 1 \mu\text{m}$ cascodes	24/4, $L = 3 \mu\text{m}$ cascodes ($W = 10L$)
bias references	90 μA	26 μA
output I_Q (P/N)	240 μA	212 μA (class-AB devices 5.2/4.8 μA)
I_{DD}	427 μA	296 μA

1.01 V. Only XFNL replicates the class-AB control. The circuit works, but the replica is not a true replica.

APPENDIX FILES

All files are in sim/:

- d2s_mpdda.spice: proposed driver, subcircuit d2s_mpdda (pins as d2s_miller; parameters wc, lc, lcas, rl).
- units.spice: DDA units unit_r, unit_r2 (used by d2s_mpdda), unit_t, unit_w, unit_q.
- d2s_bias_lp.spice: bias fixture for d2s_mpdda (pins as d2s_bias).

TABLE VIII
PERFORMANCE: FIRST VERSION VS. D2S_MPDDA

	first version	d2s_mpdda
T_0 : 1 k Ω 100 pF / 50 Ω / 100 pF	66.5/40.7/96.5 dB	78.9/53.1/104.3 dB
f_c /PM, 100 pF	2.02 MHz/66.4°	1.77 MHz/62.0°
f_c /PM, 300 pF	1.59 MHz/45.6°	1.39 MHz/42.0°
f_c /PM, 1 nF	1.00 MHz/26.5°	0.87 MHz/24.1°
gain, 11 corners	0.4870... 0.5249	0.4992... 0.5000
nonlinearity, tt / worst (ss 27 °C)	5.52/12.69 mV	1.24/3.50 mV
THD 10/100 kHz, 1 k Ω 100 pF	−51.1/ − 50.2 dB	−58.5/ − 51.3 dB
THD 10/100 kHz, 50 Ω	−46.2/ − 29.1 dB	−45.4/ − 26.9 dB
$t_1\%$, 1 k Ω 100 pF	307 ns	327 ns
offset σ / gain σ (30 seeds)	14.4 mV / 1.60 %	5.3 mV / 1.11 %
noise, 1 kHz (nV/ $\sqrt{\text{Hz}}$)	1599	676
noise, 100 kHz (nV/ $\sqrt{\text{Hz}}$)	200	152
noise, 10 Hz–10 MHz ($\mu\text{V rms}$)	278	231

TABLE IX
LOOP GAIN VS. FOLD AND MIRROR CASCODE LENGTH ($W = 10L$)

L	T_0 , 1 k Ω 100 pF	T_0 , 100 pF	T_0 , 50 Ω	PM, 100 pF	area
1 μm	69.7 dB	99.6 dB	43.9 dB	63.4°	1081 μm^2
2 μm	75.4 dB	103.1 dB	49.6 dB	62.8°	1201 μm^2
3 μm	78.9 dB	104.3 dB	53.1 dB	62.0°	1401 μm^2

- d2s_mp.spice, ccomp.spice: MP-DDA in the first version's sizing; MOM or MOS capacitors through a test-

TABLE X
IN-SLOT, LOAD-COMPENSATED DRIVER WITHOUT COMPENSATION
CAPACITOR

	uncascoded	cascoded
T_0	37.2 dB	74.0 dB
PM, 5 pF	16°	unstable
PM, 20 pF	47°	39°
PM, 100 pF	78°	77°
PM, 1 nF	89°	89°
1 k Ω 100 pF	no loop gain	no loop gain
I_{DD}	174 μ A	169 μ A

bench wrapper.

- `d2s_lc2.spice`: in-slot, load-compensated, cascoded (`d2s_lc2`) and uncascoded (`d2s_lc2_nc`).
- `run_improvements.py`: reproduces every number here; imports `../run_d2s.py` unchanged.
- `results_improvements.txt`: its output (3 min 49 s on two cores).

REFERENCES

- [1] C. H. Maier, “Two proof-of-concept class-AB analog pad drivers built on ESD-clamp output devices in IHP SG13CMOS5L,” design note `class_ab_pad_driver.md`, Sep. 2026.
- [2] R. Sarpeshkar, “Efficient precise computation with noisy components: Extrapolating from an electronic cochlea to the brain,” Ph.D. dissertation, California Inst. Technol., Pasadena, CA, USA, 1997, ch. 3.
- [3] B. A. Minch, “Analysis, synthesis, and implementation of networks of multiple-input translinear elements,” Ph.D. dissertation, California Inst. Technol., Pasadena, CA, USA, 1997.
- [4] B. A. Minch, “Multiple-input translinear element log-domain filters,” *IEEE Trans. Circuits Syst. II*, vol. 48, no. 1, pp. 29–36, Jan. 2001.
- [5] P. Hasler, Ph.D. dissertation, California Inst. Technol., Pasadena, CA, USA, ch. 4–5 (local copy: `FloatingGate/Hasler/thesis_chapter*.ps`).
- [6] E. Sánchez-Sinencio, “Floating gate techniques and applications,” lecture notes, Analog and Mixed-Signal Center, Texas A&M Univ., College Station, TX, USA.
- [7] A. J. Montalvo and J. J. Paulos, “Improved floating-gate devices using standard CMOS technology,” *IEEE Electron Device Lett.*, vol. 14, no. 8, pp. 372–374, Aug. 1993.
- [8] R. T. Edwards, Ph.D. dissertation, Dept. Elect. Comput. Eng., Johns Hopkins Univ., Baltimore, MD, USA, §§3.5–3.10 (local copy: `TimEdwards/thesis.pdf`).
- [9] R. T. Edwards and G. Cauwenberghs, “Synthesis of log-domain filters from first-order building blocks,” *Analog Integr. Circuits Signal Process.*, vol. 22, pp. 177–186, 2000.
- [10] R. T. Edwards and G. Cauwenberghs, “A second-order log-domain bandpass filter for audio frequency applications,” in *Proc. IEEE ISCAS*, Monterey, CA, USA, 1998, vol. 3.
- [11] G. Efthivoulidis and Y. Tsividis, “Signal analysis of externally linear filters,” in *Proc. IEEE ISCAS*, 1999, vol. 6, pp. 65–68.
- [12] J. Mulder, A. C. van der Woerd, W. A. Serdijn, and A. H. M. van Roermund, “General current-mode analysis method for translinear filters,” *IEEE Trans. Circuits Syst. I*, vol. 44, no. 3, pp. 193–197, Mar. 1997.
- [13] J. Mulder, A. C. van der Woerd, W. A. Serdijn, and A. H. M. van Roermund, “An RMS-DC converter based on the dynamic translinear principle,” *IEEE J. Solid-State Circuits*, vol. 32, no. 7, pp. 1146–1150, Jul. 1997.
- [14] S. Hiseni, C. Sawigun, and W. A. Serdijn, “Dynamic translinear non-linear energy operator,” in *Proc. ECCTD*, 2009, pp. 153–156.
- [15] R. J. Wiegink, *Analysis and Synthesis of MOS Translinear Circuits*. The Hague, The Netherlands (as cited in [8]).
- [16] E. Säckinger and W. Guggenbühl, “A versatile building block: The CMOS differential difference amplifier,” *IEEE J. Solid-State Circuits*, vol. SC-22, no. 2, pp. 287–294, Apr. 1987.
- [17] E. Seevinck and R. F. Wassenaar, “A versatile CMOS linear transconductor/square-law function circuit,” *IEEE J. Solid-State Circuits*, vol. SC-22, no. 3, pp. 366–377, Jun. 1987.
- [18] R. Hogervorst, J. P. Tero, R. G. H. Eschauzier, and J. H. Huijsing, “A compact power-efficient 3 V CMOS rail-to-rail input/output operational amplifier for VLSI cell libraries,” *IEEE J. Solid-State Circuits*, vol. 29, no. 12, pp. 1505–1513, Dec. 1994.
- [19] J. M. Monticelli, “A quad CMOS single-supply op amp with rail-to-rail output swing,” *IEEE J. Solid-State Circuits*, vol. SC-21, no. 6, pp. 1026–1034, Dec. 1986.
- [20] B. K. Ahuja, “An improved frequency compensation technique for CMOS operational amplifiers,” *IEEE J. Solid-State Circuits*, vol. SC-18, no. 6, pp. 629–633, Dec. 1983 (from memory, not re-checked).